Youhao **Hu**

1st Duxue Road, Dongchong Town, Guangzhou, China

(+86) 18255189791 | yhu543@connect.hkust-gz.edu.cn | https://thingsc.github.io/youhaohu-HKUST-CV/

Summary

I am a Ph.D. candidate in Power Electronics at The Hong Kong University of Science and Technology (Guangzhou), supervised by Dr. Wei Han and Prof. C. C. Chan. My research interests lie in power electronics, wireless power transfer (WPT), motor drive systems, and advanced control techniques for energy conversion applications.

My work combines theoretical research with practical engineering development, spanning system modeling, control algorithm design, embedded implementation, hardware prototyping, and experimental validation. Recent research topics include wireless motor systems, high-efficiency WPT systems, adaptive and sliding mode control, and wireless-powered propulsion systems for unmanned vessels.

I am experienced in MATLAB/Simulink, Python, and C, with strong expertise in automatic code generation, real-time control systems, and embedded deployment. My goal is to bridge advanced control theory and industrial applications by developing efficient, reliable, and intelligent energy conversion systems.

Area of Expertise

Power Electronics and Wireless Power Transfer Systems

Experienced in the modeling, analysis, and control-oriented design of power electronic systems, wireless power transfer (WPT) systems, and bidirectional wireless motor systems. Familiar with system-level design, parameter optimization, and experimental validation.

Complex Dynamical System Modeling

Experienced in modeling and analyzing complex dynamical systems, including power electronic converters, wireless power transfer systems, motor drive systems, and multi-physics coupled systems. Capable of conducting system-level modeling, stability analysis, and controller design.

Advanced Control Theory and Controller Design

Proficient in advanced control methodologies including Sliding Mode Control (SMC), Adaptive Control, and Extended State Observer (ESO)-based control. Experienced in nonlinear controller design, signal processing, and real-time control implementation.

Embedded Control System Development

Experienced in the complete development cycle of embedded control systems, from control algorithm design and simulation to real-time deployment on embedded platforms. Proficient in MATLAB/Simulink, Python, and C programming.

Intelligent Control and Machine Learning

Familiar with machine learning algorithms and neural network techniques, with experience in NN-assisted control and hybrid intelligent control strategies that integrate data-driven methods with conventional control approaches.

Education & Experience

The Hong Kong University of Science and Technology — *Guangzhou & Hong Kong*

PH.D. IN POWER ELECTRONICS — *Sept. 2021 - Present*

- Ph.D. Candidate — HKUST (Guangzhou) — *Sept. 2022 - Present*
- Cross-Campus-Study Exchange Ph.D. Programme — HKUST, Hong Kong — *Feb. 2025 - Aug. 2025*
- Research Assistant — HKUST (Guangzhou) — *Sept. 2021 - Aug. 2022*

Hefei University of Technology — *Hefei, China*

M.S. IN CONTROL ENGINEERING — *Sept. 2017 - Jan. 2021*

- Research Assistant — *Sept. 2020 - Jan. 2021*
- M.S. — *Sept. 2017 - Jun. 2020*

Shandong University of Technology — *Zibo, China*

B.S. IN AUTOMATION — *Sept. 2013 - Jun. 2017*

Publication

SELECTED JOURNALS

1 **Power Boosting of Strongly Coupled Wireless Power Transfer Systems by Aligning 1st and 3rd Harmonic Frequencies with Splitting Frequencies** — *IEEE Transactions on Power Electronics 2025*

BOWANG ZHANG, **YOUHAO HU**, WEIKANG HU AND WEI HAN

(DOI: 10.1109/TPEL.2025.3544291)

2 **Maximizing Wireless Power Transfer Efficiency at Exceptional Points** *Communications Engineering 2025*

Weikang Hu, Bowang Zhang, **Youhao Hu**, Haoxiang Li and Wei Han

(DOI: 10.1038/s44172-025-00445-y)

3 **An Active-Rectifier Wireless Motor System with Dual-Side Phase Shift Control for Phase Synchronization and Efficiency Optimization** *IEEE Transactions on Power Electronics 2024*

Youhao Hu, Wei Han, Bowang Zhang and Weikang Hu

(DOI: 10.1109/TPEL.2024.3493093)

4 **Fixed-Time Adaptive Sliding Mode Control for Vehicular Electronic Throttle with Actuator Saturation Using Extreme Learning Machine** *IEEE Transactions on Vehicular Technology 2024*

Youhao Hu, Hai Wang, Amirmehdi Yazdani, Liqing Chen, Ming Yu, Jinchuan Zheng and Zhihong Man

(DOI: 10.1109/TVT.2024.3438802)

5 **ELM-Based Adaptive Practical Fixed-Time Voltage Regulation in Wireless Power Transfer System** *Energies 2023*

Youhao Hu, Bowang Zhang, Weikang Hu and Wei Han

(DOI: 10.3390/en16031016)

6 **Adaptive Tracking Control of an Electronic Throttle Valve Based on Recursive Terminal Sliding Mode** *IEEE Transactions on Vehicular Technology 2021*

Youhao Hu, Hai Wang, Shuping He, Jinchuan Zheng and Zhaowu Ping

(DOI: 10.1109/TVT.2020.3045778)

7 **Robust Tracking Control for Vehicle Electronic Throttle Using Adaptive Dynamic Sliding Mode and Extended State Observer** *Mechanical Systems and Signal Processing 2020*

Youhao Hu and Hai Wang

(DOI: 10.1016/j.ymssp.2019.106375)

8 **Adaptive Full-Order Sliding Mode Control for Electronic Throttle Valve System with Fixed-Time Convergence Using Extreme Learning Machine** *Neural Computing and Applications 2021*

Youhao Hu, Hai Wang, Amirmehdi Yazdani and Zhihong Man

(DOI: 10.1007/s00521-021-06365-0)

9 **Extreme-Learning-Machine-Based FNTSM Control Strategy for Electronic Throttle** *Neural Computing and Applications 2020*

Youhao Hu, Hai Wang, Zhenwei Cao, Jinchuan Zheng, Zhaowu Ping, Long Chen and Xiaozheng Jin

(DOI: 10.1007/s00521-019-04446-9)

Conference Papers

1 **A Closed-Loop-Controlled Wireless Motor System with Maximum Efficiency Point Tracking** *IEEE Energy Conversion Congress and Exposition (ECCE) 2024*

Youhao Hu, Wei Han, Bowang Zhang and Weikang Hu

(DOI: 10.1109/ECCE55643.2024.10861822)

2 **Robust Speed Control for Wireless Motor Systems with Maximum Efficiency Point Tracking** *IEEE Annual Conference of the Industrial Electronics Society (IECON) 2023*

Youhao Hu, Bowang Zhang, Weikang Hu and Wei Han

(DOI: 10.1109/IECON51785.2023.10312152)

3 **Robust Fast Nonsingular Terminal Sliding Mode Control Strategy for Electronic Throttle Based on Extreme Learning Machine** *Chinese Control Conference (CCC) 2019*

Youhao Hu, Hai Wang, Zhenwei Cao, Zhihong Man, Ming Yu and Zhaowu Ping

(DOI: 10.23919/CHICC.2019.8866147)

4 **Discrete-Time Fast Sliding-Mode Control for Permanent Magnet Linear Motor with Nonlinear Extended State Observer** *International Conference on Power and Energy Systems (ICPES) 2019*

Youhao Hu and Hai Wang

(DOI: 10.1109/ICPES47639.2019.9105583)

Book Chapters

1 **Real-Time Control Systems with Applications in Mechatronics** *Handbook of Real-Time Computing*

Hai Wang, **Youhao Hu**, Mao Ye, Jie Zhang, Zhenwei Cao, Jinchuan Zheng and Zhihong Man

(DOI: 10.1007/978-981-4585-87-3_41-1)

2 **Challenge and Trend on Energy Digitalization** *Integration of Energy, Information, Transportation and Humanity*

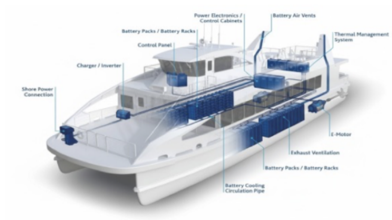
Wei Han, C. C. Chan, **Youhao Hu**, Chang Liu and George You Zhou

(DOI: 10.1016/B978-0-323-95521-8.00006-3)

Honors & Awards

2026	**Gold Medal, International Exhibition of Inventions Geneva**	*Palexpo SA, CH*
2022	**Fully Funded PhD Scholarship**	*HKUST(GZ), CN*
2020	**Outstanding Graduate of Anhui Province**	*Anhui Province, CN*
2019	**National Graduate Scholarship**	*MOE, CN*
2016	**Meritorious Winner, Interdisciplinary Contest in Modeling (ICM)**	*COMAP, US*
2016–2019	**Multiple Academic Excellence Scholarships and Honors**	*HFUT & SDUT, CN*

New Propulsion System for Unmanned Vessels: Wireless-Powered Full- Azimuth Podded Thruster

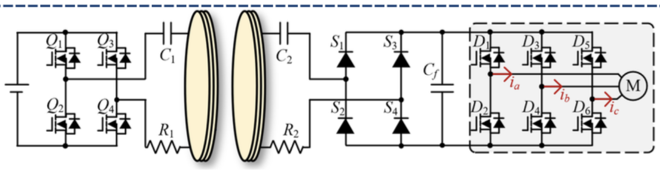


Key issues

Conventional podded propulsion solutions suffer from sealing reliability and high maintenance costs.

My Responsibilities

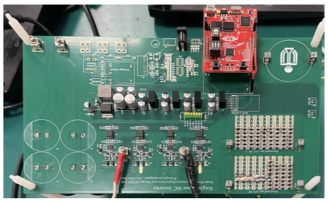
- Wireless power transfer system modeling and parameter design
- High-frequency wireless power converter control development and system integration
- Motor drive control development (FOC / real-time control)
- Sealed mechanical structure design and 3D modeling
- Prototype system integration, debugging, and experimental validation



A. System topology

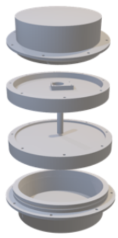
- Series-series compensated wireless power transfer

GaN motor driver



SiC high frequency inverter

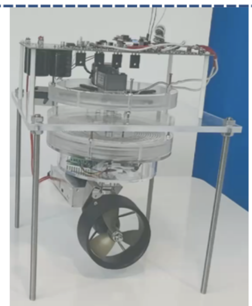
B. In-House Developed Hardware



C. 3D structure modelling

Key Specifications

- 1200 W / 100 V GaN motor driver
- 5 kW integrated wireless power transfer inverter
- Power transfer distance: 5–10 cm
- Peak wireless power transfer efficiency: 92%
- Structural design for fully sealed applications

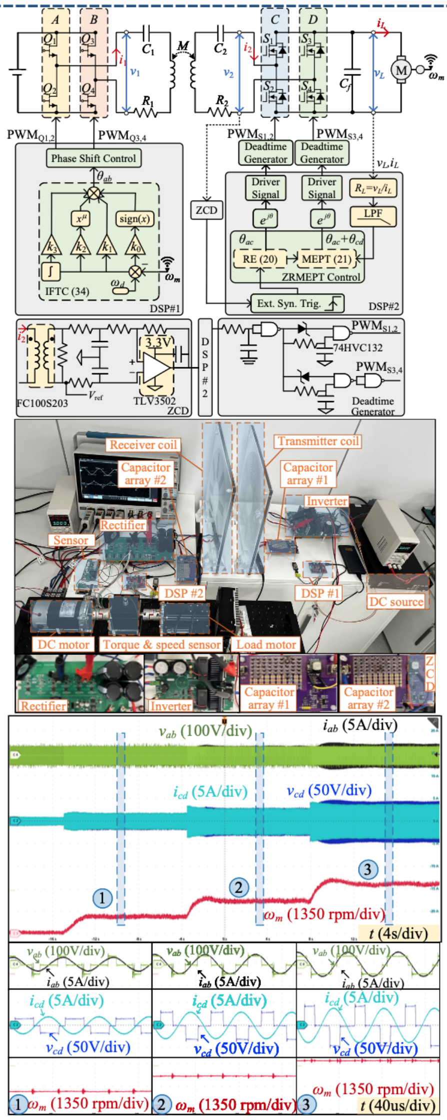


Prototype

An Active-Rectifier Wireless Motor System with Dual-Side Phase Shift Control for Phase Synchronization and Efficiency Optimization

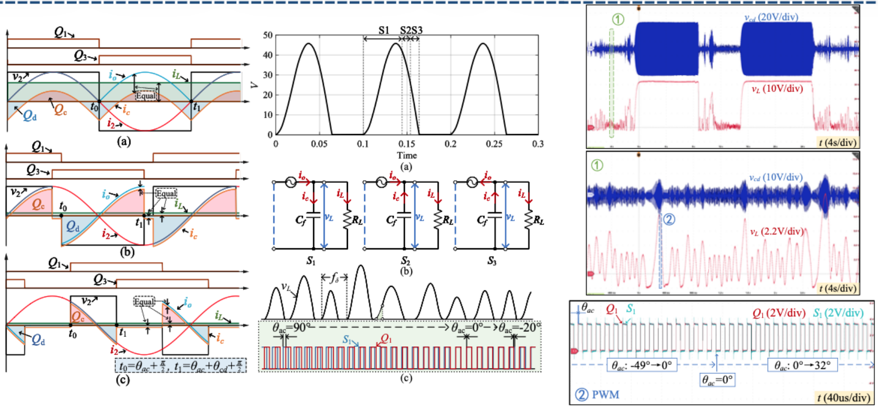
Published in *IEEE Transactions on Power Electronics*

This paper introduces a novel active-rectifier wireless motor (AR-WM) system with dual-side phase shift control (DS-PSC) to not only synchronize dual-side phases eliminating reactive power but also maximize the system efficiency. First, by detecting the zero-crossing point of the secondary resonant current, the phase angle between two converters on both sides is locked at 90 degree to maintain the real-time frequency synchronization. Second, the primary-side PSC determines the load voltage for regulating the motor speed, while the secondary-side PSC is responsible for achieving dual-side zero phase angle by synchronizing dual-side phases and tracking the maximum efficiency point by rapidly tuning the equivalent impedance. Besides, to improve the dynamic performance of the AR-WM system, an integral finite-time control approach is utilized, resulting in a 43.89% reduction in speed response time compared to the traditional PI control method. Finally, a 180 W experimental prototype is built with a maximum system efficiency of 92.37% at a transfer distance of 10 cm.



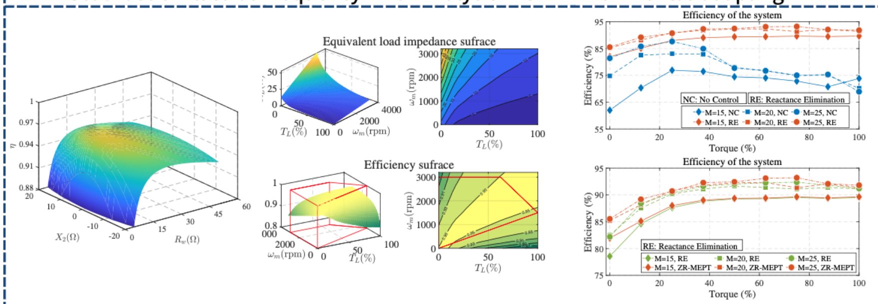
High efficiency and robust wireless motor system

A. Frequency synchronization via zero crossing detection.
B. Maximum efficiency tracking via zero-reactance MEPT.
C. Robust fast speed response via finite-time controller.



A. Dual-side random frequency analysis and elimination

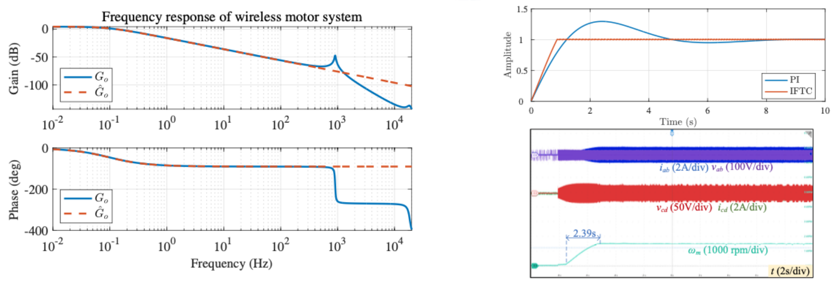
- Stochastic Bilateral Frequency Error Analysis with Environmental Coupling



B. Zero-reactance MEPT

- Reactance elimination via dual-side phase shift control.
- Maximum efficiency point tracking via active rectifier.
- High efficiency in full range of operation of the motor.

92.37% efficiency in distance of 10 cm for wireless motor system



C. System dynamic simplification and finite-time control design

- System order reduction via dominant pole analysis.
- Finite-time controller using variable structure control action.
- Invariance to disturbance and no overshoot exists.

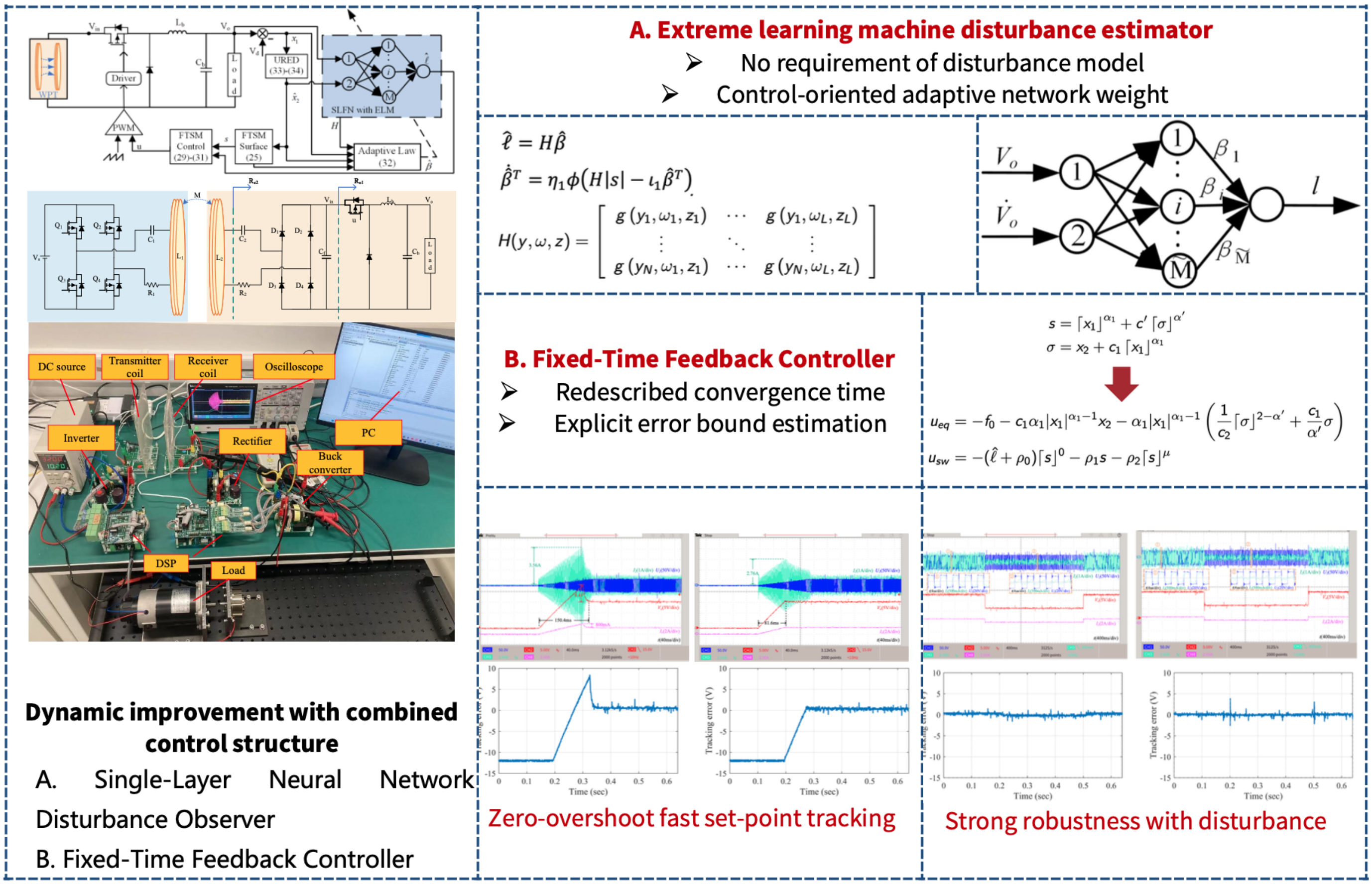
43.89% overshoot reduction compared with PI control

DOI: 10.1109/TPEL.2024.3493093

ELM-Based Adaptive Practical Fixed-Time Voltage Regulation in Wireless Power Transfer System

Published in *Energies*

This paper proposes an extreme learning machine (ELM)-based adaptive sliding mode control strategy for the receiver-side buck converter system in the wireless power transfer system subjecting to the lumped uncertainty. The proposed control strategy utilizes a singularity-free fixed-time sliding mode (FTSM) feedback control, which ensures a fixed-time convergence for both the sliding variable and voltage tracking error. An ELM-based uncertainty bound estimator is further designed to learn the uncertainty bound information in real-time, which opportunely loosens the constraint of bound information requirement for sliding mode control design. The global stability of the closed-loop system is rigidly analyzed, and the good performance of the proposed control strategy is validated by comparison experiments which exhibit ideal overshoot elimination, 45.70-51.72% reduction of settling time, and 13.65-36.96% reduction of the root mean square value for voltage tracking error with respect to different load types.

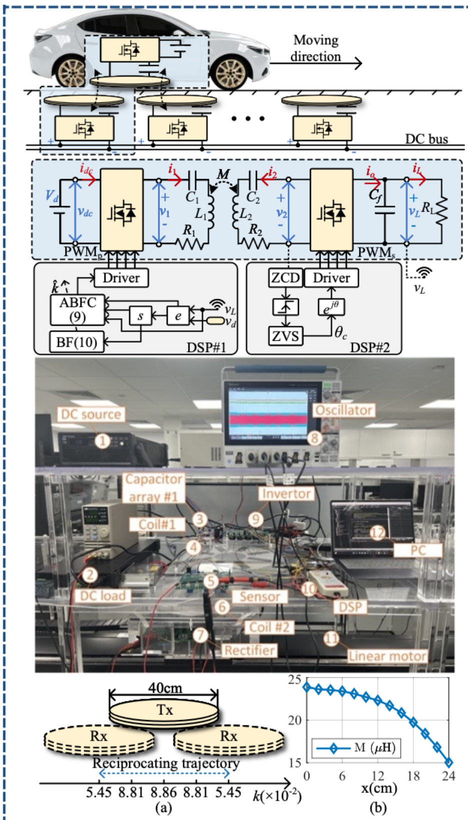


DOI: 10.3390/en16031016

Adaptive Barrier-Function Sliding Mode Control for Voltage Stabilization in Dynamic Wireless Power Transfer System

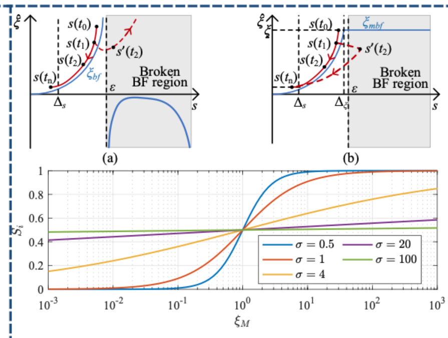
Under Review

This paper proposes an adaptive barrier-function sliding-mode control (ABF-SMC) strategy that leverages an exponential-power barrier function to adjust the control gain dynamically. Without requiring prior knowledge of disturbance magnitudes, the ABF-SMC simultaneously stabilizes the receiver voltage within a predefined region and suppresses fluctuations in the transmitter input current, while ensuring a continuous, non-overestimated control action. Experiments on a DWPT prototype traveling at up to 10 km/h show that, compared with proportional-integral (PI) control and adaptive sliding-mode control (A-SMC), the proposed method reduces receiver-voltage ripple by 26.63% and 5.19%, and suppresses input-current overshoot by 10.15% and 3.19%, respectively.



ABF-SMC for DWPT system

- Adaptive barrier function-based sliding mode control.
- Real-time frequency synchronization and soft switching.



System model:

$$\dot{v}_L = -\frac{1}{\tau_L} v_L + \frac{8R_L}{\pi^2 \omega M_0 \tau_L} V_d + f_M$$

ABF-SMC:

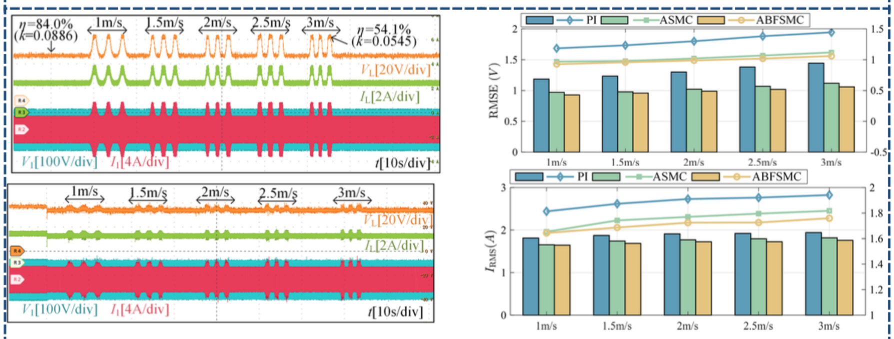
$$u = \frac{1}{\tau_L} v_L - \gamma \lceil e_v \rfloor^{\mu} - \xi_1 s - \hat{\xi} \lceil s \rfloor^{0}$$

$$\hat{\xi} = \begin{cases} \int_0^{\tau} \vartheta |s| d\tau, & 0 < t < t_0 \\ \xi_{bf1}(s) + \xi_{bf2}(s), & t \geq t_0 \end{cases}$$

$$\xi_{bfi}(s) = \frac{|s|^{m_i}}{\lceil \varepsilon - |s| \rfloor^{m_i}}$$

A. Controller design

- Adaptive gain with no overestimation
- Saturation modification for the sampled system
- Faster convergence with an exponential power term



B. Experimental verification

- 26% reduction in RMS error for voltage tracking
- 10% reduction in the current stress of the primary inverter
- Maximum efficiency 84% with k=0.0886
- Robust to speed variation, better performance with faster speed

Adaptive Finite-/Fixed-time Sliding Mode Control with Disturbance Observers for Electronic Vehicle Throttle System

Published in *IEEE Transactions on Vehicular Technology*, *Neural Computing and Application*, *Mechanical Systems and Signal Processing*

This project studies finite-time and fixed-time sliding mode control for electronic throttle systems with parametric uncertainties, lumped uncertainty, and actuator constraints. The feedback controller adopts fixed-time sliding mode (FTSM) [1,5], recursive terminal sliding mode (RTSM) [2], dynamic sliding mode (DSM) [3], and nonsingular terminal sliding mode (NTSM) [4] controller to ensure the rapid and robust control property. For meeting requirements in all operation conditions, parameters adaptive mechanism and disturbance estimation methods are further developed to tune the parameters online [1-3] and compensate for the lumped uncertainty in real time [1-5].

The project [1] adopts a nonsingular fixed-time sliding mode to realize fixed-time tracking of throttle opening, an extreme learning machine to estimate the system uncertainty, an auxiliary system to encounter the actuator saturation, and a direct adaptive technique to estimate controller parameters in real time. The results were published in *IEEE Transactions on Vehicular Technology*.

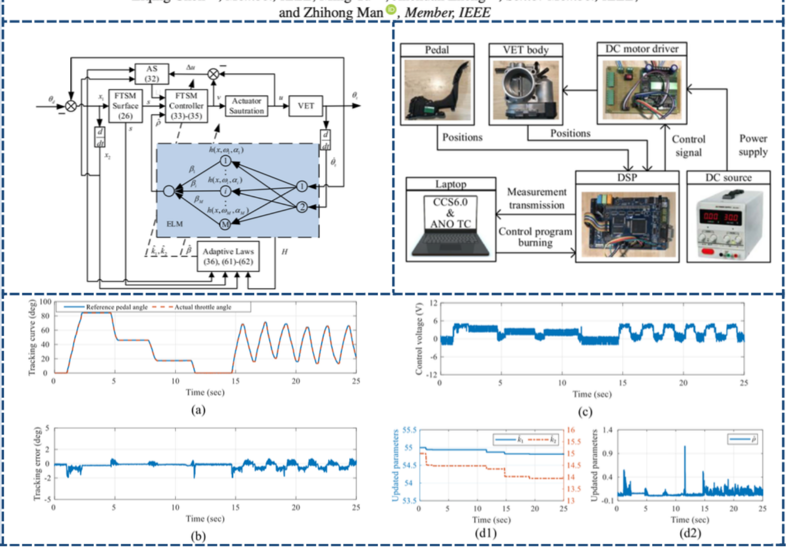
Highlights:

- Nonsingular fixed-time sliding mode controller with adaptive gains
- ELM-based disturbance estimation with online learning weights.
- Indicating and attenuating saturation effect via an auxiliary system (AS).
- All variables including sliding variables, AS variables, adaptive weights, and system states converge in a uniformly bounded convergence time.

IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, VOL. 73, NO. 12, DECEMBER 2024 18483

Fixed-Time Adaptive Sliding Mode Control for Vehicular Electronic Throttle With Actuator Saturation Using Extreme Learning Machine

Youhao Hu, Hai Wang, *Senior Member, IEEE*, Amirmehdi Yazdani, *Senior Member, IEEE*, Liqing Chen, *Member, IEEE*, Ming Yu, Jinchuan Zheng, *Senior Member, IEEE*, and Zhihong Man, *Member, IEEE*



DOI: 10.1109/TVT.2024.3438802

The project [2] adopts a recursive terminal sliding mode to realize finite time tracking of throttle opening, adopts direct adaptive technique to estimate the upper bound of system disturbance and controller parameters in real time. The results were published in *IEEE Transactions on Vehicular Technology*.

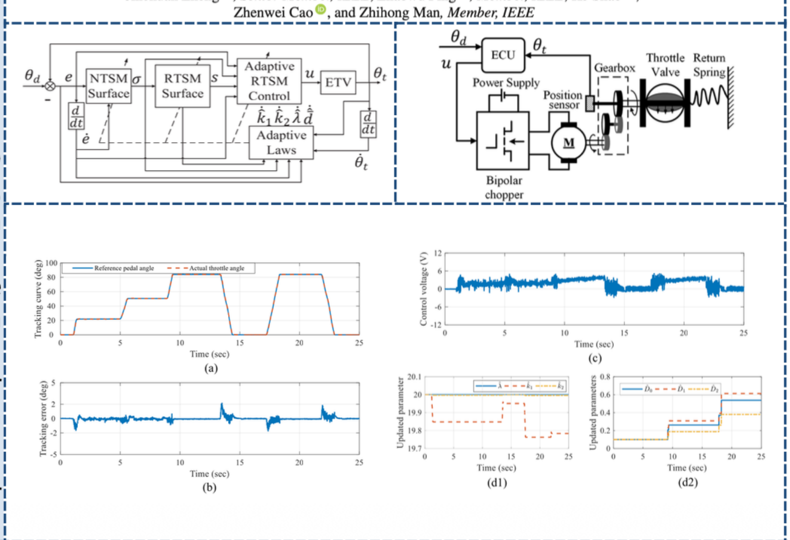
Highlights:

- **Highly cited paper in 2021**
- The recursive terminal sliding mode is used to realize the finite time convergence of output error, sliding mode variable, and adaptive estimation variable.
- The integral term is introduced to eliminate the reaching phase of sliding mode control which improves the robustness.
- Adaptive estimation method of disturbance upper bound based on property of bounded system disturbance.
- Online adaptive adjustment of controller parameters.

IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, VOL. 70, NO. 1, JANUARY 2021 251

Adaptive Tracking Control of an Electronic Throttle Valve Based on Recursive Terminal Sliding Mode

Youhao Hu, Hai Wang, *Senior Member, IEEE*, Shuping He, *Senior Member, IEEE*, Jinchuan Zheng, *Senior Member, IEEE*, Zhaowu Ping, *Member, IEEE*, Ke Shao, Zhenwei Cao, and Zhihong Man, *Member, IEEE*



DOI: 10.1109/TVT.2020.3045778

In this project [4,5], the finite time/fixed time sliding mode control is used to realize the accurate tracking of throttle opening, and an extreme learning machine (ELM) is used to learn the upper bound information of system disturbance in real time, so as to enhance the robustness of the system. The results have been published in *Neural Computing and Applications*

Highlights:

- Finite time / fixed time sliding mode feedback control
- Real-time adaptive learning disturbance / upper bound based on ELM networks
- Global finite time / fixed time stability

Neural Computing and Applications
https://doi.org/10.1007/s00521-019-04446-9

EXTREME LEARNING MACHINE AND DEEP LEARNING NETWORKS

Extreme-learning-machine-based FNTSM control strategy for electronic throttle

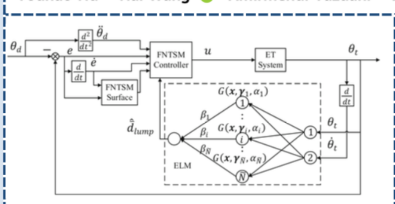
Youhao Hu[1] · Hai Wang[2] · Zhenwei Cao[3] · Jinchuan Zheng[3] · Zhaowu Ping[1] · Long Chen[4] · Xiaozheng Jin[1]

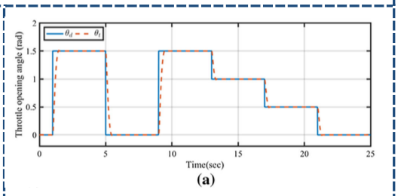
Neural Computing and Applications
https://doi.org/10.1007/s00521-021-06365-0

S.I: COMPUTATIONAL INTELLIGENCE-BASED CONTROL AND ESTIMATION IN MECHATRONIC SYSTEMS

Adaptive full order sliding mode control for electronic throttle valve system with fixed time convergence using extreme learning machine

Youhao Hu[1] · Hai Wang[1] · Amirmehdi Yazdani[1] · Zhihong Man[2]





DOI: 10.1007/s00521-021-06365-0
DOI: 10.1007/s00521-019-04446-9

In the project [3], a dynamic sliding mode control is utilized to realize the accurate tracking of throttle opening angle, an improved extended state observer is designed to estimate the system disturbance in real time, and the direct adaptive technology is adopted to eliminate the observer error and enhance the robustness of the system. The results are published in *Mechanical System and Signal Processing*

Highlights:

- Dynamic sliding mode feedback control
- Improved extended state observer design for discontinuous dynamics
- Global stability of systems with observer dynamics
- Adaptive observation error suppression

Contents lists available at ScienceDirect

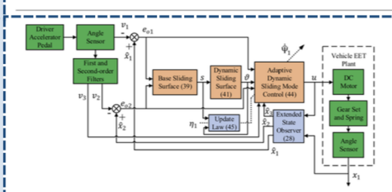
Mechanical Systems and Signal Processing

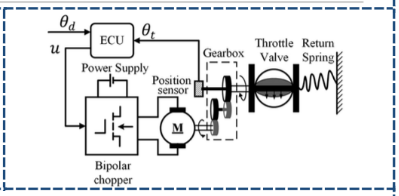
journal homepage: www.elsevier.com/locate/ymssp

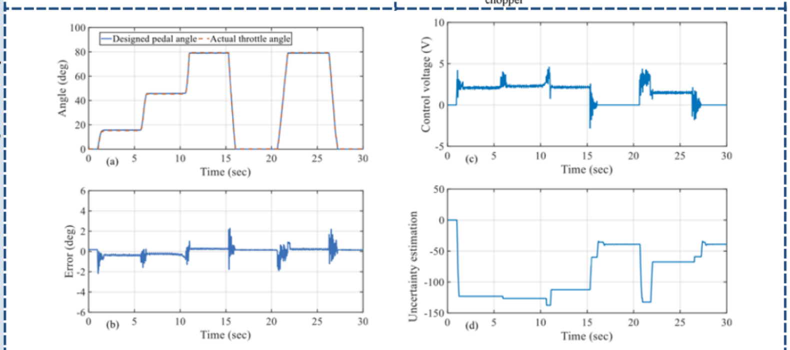
Robust tracking control for vehicle electronic throttle using adaptive dynamic sliding mode and extended state observer

Youhao Hu[a], Hai Wang[b,*]

[a] School of Electrical and Automation Engineering, Hefei University of Technology, Hefei 230021, China
[b] College of Science, Health, Engineering, and Education, Murdoch University, Perth WA6150, Australia







DOI: 10.1016/j.ymssp.2019.106375